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Reference

video

book

OpenCourseWare - Mathematics for Computer Science - 4. Number Theory 2

Lecture Note

5. Number Theory II

Encryption

before hand: "keys" are exchanged

encryption : $m' = E_{key}(m)$

decryption : $m = D_{key}(m')$

Turing' code V1

victory

$m = 220903201518\,2513$ ← prime number

Before hand: exchange secrect prime, k

Enc: $m' = mk$ Hard to factor a product of two large prime

Dec: $m'/k = m$

$m_1' = m_1 k$ not secure we get k

$m_2' = m_2 k$

$\gcd(m_1', m_2') = k$

Turing Code V2

Before hand: exchange a pubic prime p & a screct prime k

$\{0, 1, \ldots, p-1\}$

message : a number, $m \in [$

Encryption : $m' = \text{rem}(mk, p)$

Def: a, b relatively prime iff $\gcd(a,b) = 1$ iff $\exists_{s,t}\ sa + tb = 1$

Def: x is congruence to y modulo n: $x \equiv y \pmod{n}$ iff $n \mid (x-y)$

Ex $31 \equiv 16 \pmod 5$ both x & y have same "rem" when divided by n

Def: the multiplicative inverse of x mod n is a number $x^{-1} \in [0, 1, \cdots, n-1]$

s.t $x \cdot x^{-1} \equiv 1 \pmod n \rightarrow x \cdot x^{-1} \bmod n = 1$

Ex $2 \cdot 3 \equiv 1 \pmod 5$

$2 \equiv 3^{-1} \pmod 5$

$3 \equiv 2^{-1} \pmod 5$

$\underbrace{5 \cdot 5}_{25} = 1 \pmod 6$

$5 = 5^{-1} \pmod 6$

3 is modular inverse of 2 for n=5

multiplicative inverse

multiplicative inverse of x

the whole number that multiplies with x such that the result, when divided by n, leave the reminder of 1.

Decryption:

$$\underbrace{rem(mk, p)}_{m'} \equiv mk \pmod{p}$$

If $k.k^{-1} \equiv 1 \pmod{p}$ then $m'.k^{-1} \equiv mk.k^{-1} \pmod{p}$

$$m'.k^{-1} \equiv m \pmod{p}$$

$$\downarrow$$

$$\in \{0, 1, \ldots, p-1\}$$

$$m = rem(m'.k^{-1}, p)$$

Known-plaintext attack: breaking 2nd schema

known message, m & encryption $m' = rem(mk, p)$ & p

$$m' \equiv mk \pmod{p}$$

$$\gcd(m, p) = 1$$

compute m^{-1} s.t $m.m^{-1} \equiv 1 \pmod{p}$

$m' \equiv mk \pmod p$

$m' \cdot m^{-1} \equiv \underbrace{m \cdot m^{-1}}_{1} \cdot k \equiv k \pmod p$

now you now k, compute k^{-1} from $k \cdot k^{-1} = 1 \pmod p$

for future message, we can use k^{-1} to decrypt

We have talked

1. Encryption
   a) Turing code v1
   b) Turing code v2
2. Modular arthemetic
   a) $x \equiv y \pmod n$
   b) multiplication inverse

we will talk

3. Euler's totient function
   a) Euler's theorem
   b) Fermat's (little) theorm
4. Encryption
   a) RSA
   b) FHH over the integers

Euler's totient function

Def: $\phi(n)$ denotes numbers of integers in $\{1,2,3,\dots,n-1\}$ that are relatively prime to n

Ex: $n=12$, $\phi(12)=4$ $\{1,5,7,11\}$

$n=15$, $\phi(15)=8$ $\{1,2,4,7,8,11,13,14\}$

Euler's theorem: if $\gcd(n,k)=1 \Rightarrow k^{\phi(n)} \equiv 1 \pmod n$

Lemma 1: If $\gcd(n,k)=1$, then $ak \equiv bk \pmod n \Rightarrow a \equiv b \pmod n$

$\gcd(n,k)=1$ iff k has a multiplication inverse

Pf: $\gcd(n,k)=1 \Leftrightarrow \exists_{s,t}\ n\cdot s+k\cdot t=1 \Leftrightarrow \exists_t\ n \mid (1-k\cdot t)$

$\Leftrightarrow \exists_t\ n \mid (kt-1)$ ↙ $5 \mid 10$ & -10

$\Leftrightarrow kt \equiv 1 \pmod n$

now t is mutiplication of k

pf: $\gcd(n,k)=1 \Rightarrow kt \equiv 1 \pmod n$

$ak \equiv bk \pmod n$

$ak\cdot t \equiv b\cdot k\cdot t \pmod n$

$a \equiv b \pmod n$

Lemma 2: If $\gcd(n,k)=1$, let $k_1,\dots,k_r$ in $\{1,2,\dots,n-1\}$ denote the

the integer relatively prime to n $(r = \phi(n))$

Then $\{rem(k_i \cdot k, n), \ldots, rem(k_r \cdot k, n)\} = \{k_1, \ldots, k_r\}$

(1) $\#$ = r (2)

Pf(1): Assume $rem(k_i k, n) = rem(k_j k, n)$

$k_i k \equiv k_j k \pmod{n}$

$k_i \equiv k_j \pmod{n}$ by Lemma 1

$k_i = k_j$

therefore $\{rem(k_1 k, n), \ldots, rem(k_r \cdot k, n)\}$ set is all different (they will only be same iff $k_1 = k_r$) has $\#$ of r

pf ②:

$\gcd(n, \text{rem}(k_i k, n)) = \gcd(n, k_i k) = 1$

$\gcd(n, k) = 1$
$\gcd(n, k_i) = 1$ } ↑

so $\gcd(n, k_i k)$ will be: subset or equal to of $\{k_1, \ldots k^n\}$ since they are relatively prime to n

from pf ① & pf ② we show lemma 2 is True

pf (Euler's Theorm):

$k_1 \times k_2 \times \ldots \times k_r = \text{rem}(k_1 k, n) \times \ldots \times \text{rem}(k_r \cdot k, n)$

$\equiv k_1 k \times k_2 k \times \ldots \times k_r \cdot k \pmod{n}$

$\equiv k_1 \times \ldots \times k_2 \times k^r \pmod{n}$

if $\gcd(k_1, n), \ldots, \gcd(k_r, n) = 1 \Rightarrow \gcd(k_1 * \ldots * k_2, n) = 1$

Assume $k_1 \times \ldots \times k_r = t$

$1 \times t \equiv t \times k^r \pmod{n}$

by lemma 1, $1 \equiv k^r \pmod{n}$ $(r = \phi(n))$ ✓

[illegible] (Little) Theorem

Fermat's little theorem

Suppose p is prime, and $k \in \{1, 2, \ldots, p-1\}$ then $k^{p-1} \equiv 1 \pmod{p}$

Pf: $1, 2, \ldots, p-1$ are relatively prime to $p \Rightarrow \phi(p) = p-1 = lc$

using Euler's th: $k^{\phi(p)} \equiv 1 \pmod{p}$ since Euler's th work for all number, change $p \to lc$

$k^{p-1} \equiv 1 \pmod{p}$

Date

RSA

beforehand : receiver create public key & secret key

1. generate two distinct prime (p, q)
2. let $n = p.q$ $(m \in \{0, \ldots, n-1\})$

3. select integer, e s.t $\gcd(e, (p-1)(q-1)) = 1$

   public key = (e, n)

4. compute [redacted], d s.t $d.e \equiv 1 \pmod{(p-1)(q-1)}$

   screct key = (d, n)

Encryption : $m' = \text{rem}(m^e, n)$

Decryption : $m = rem(m'^{d}, n)$

$m' = rem(m^e, n) \equiv m^e \pmod{n} \Rightarrow (m')^d \equiv m^{ed} \pmod{n}$

$\exists r \; ed = 1 + r(p-1)(q-1)$ from step 9.

So, $(m')^d \equiv m^{ed} \equiv m \cdot m^{r(p-1)(q-1)} \pmod{n}$

if $m \not\equiv 0 \pmod{p}$ then $m^{p-1} \equiv 1 \pmod{p}$

$\gcd(m, p) = 1$ use fermat little theorm

No.

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Fold

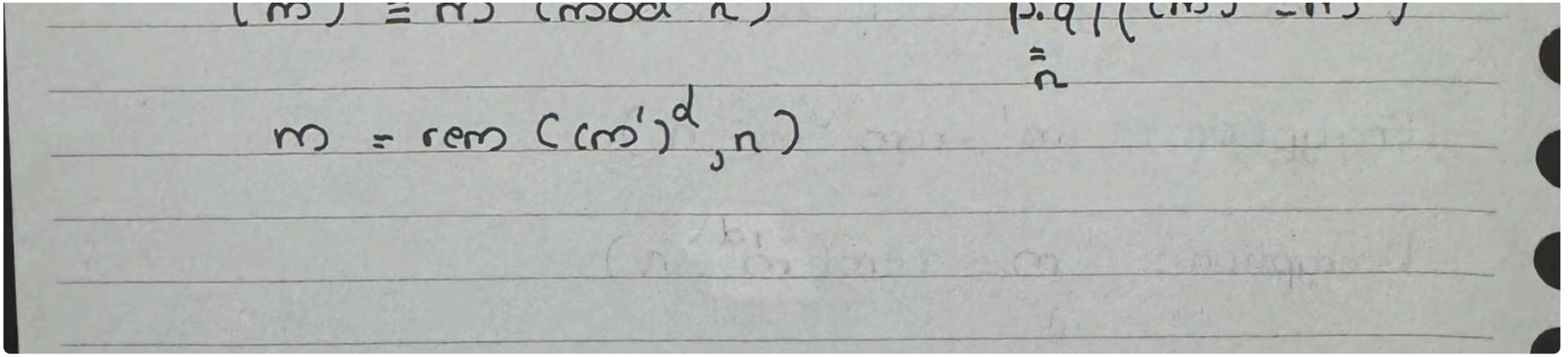
If $m \not\equiv 0 \pmod{q}$ then $m^{q-1} \equiv 1 \pmod{q}$

if $(m')^d \equiv m^{ed} \equiv m.m^{r(p-r)(q-r)} \pmod{n}$

since $n = pq$

$$(m')^d \equiv m^{ed} \equiv m.m^{r(p-r)(q-r)} \pmod{p}$$
$$(m')^d \equiv m^{ed} \equiv m.m^{r(p-r)(q-r)} \pmod{q}$$

so $(m')^d \equiv m \pmod{p}$ $\Big\}$ $p \mid ((m')^d - m)$
$(m')^d \equiv m \pmod{q}$ $\Big\}$ $q \mid ((m')^d - m)$ $\Big\}$



Book Reading : 4 Number Theory

4.5 Modular Arithmetic

a is congruent to b *modulo* n iff $n|(a-b)$ which be written as

$$a \equiv \ b \ (\text{mod}\ n)$$

example : $29 \equiv \ 15 \ (\text{mod}\ 7)$

Lemma 4.5.1 (Congruences and Remainders).

$$a \equiv b \pmod{n} \iff \text{rem}(a,n) = \text{rem}(b,n)$$

Thm $a \equiv b \pmod{n}$ iff $\text{rem}(a,n) = \text{rem}(b,n)$

proof: By the division Theorem, we can rewrite a & b

$a = q_1 n + r_1$ where $0 \leq r_1 < n$

$b = q_2 n + r_2$ where $0 \leq r_2 < n$

$a - b = (q_1 - q_2)n + (r_1 - r_2)$ where $-n < r_1 - r_2 < n$

we know $n \mid a-b$ from $a \equiv b \pmod{n} \iff n \mid a-b$

therefore $n \mid (q_1 - q_2)n + (r_1 - r_2)$

$q_1 - q_2$ is already muliple of n therefore $(r_1 - r_2)$ also must be multiple of n

given the boundry of $r_1 - r_2$ $(-n < r_1 - r_2 < n) = 0$

$r_1 = r_2$

$\text{rem}(a,n) = \text{rem}(b,n)$

example

$$29 \equiv 15 \pmod{7} \iff \operatorname{rem}(29,7) = \operatorname{rem}(17,7)$$

Corollary 4.5.2

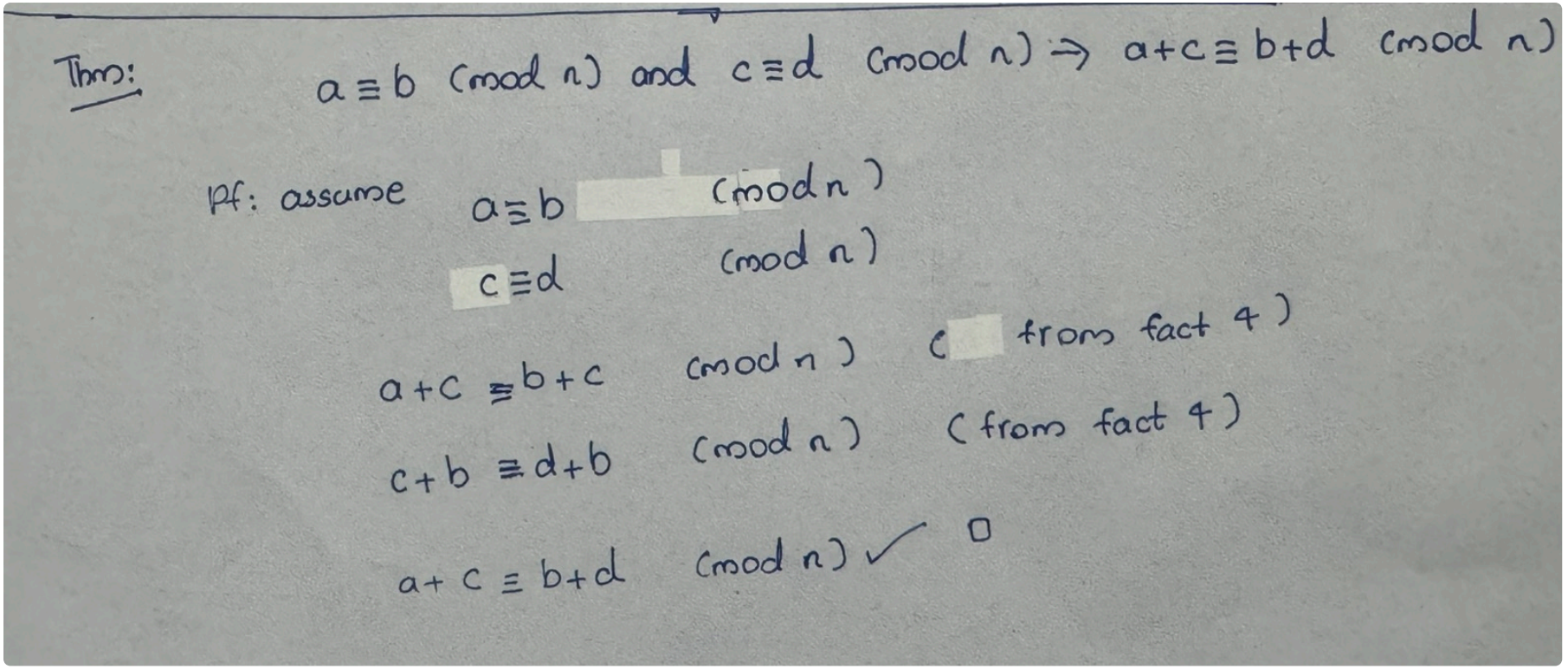
$$a \equiv \operatorname{rem}(a,n) \pmod{n}$$

Lemma 4.5.3 (Facts About Congruences)

congruences work a lot like equations, though there are a couple of exceptions

Lemma 4.5.3 (Facts About Congruences)**.** *The following hold for* $n \geq 1$*:*

1. $a \equiv a \pmod{n}$
2. $a \equiv b \pmod{n}$ *implies* $b \equiv a \pmod{n}$
3. $a \equiv b \pmod{n}$ *and* $b \equiv c \pmod{n}$ *implies* $a \equiv c \pmod{n}$
4. $a \equiv b \pmod{n}$ *implies* $a + c \equiv b + c \pmod{n}$
5. $a \equiv b \pmod{n}$ *implies* $ac \equiv bc \pmod{n}$
6. $a \equiv b \pmod{n}$ *and* $c \equiv d \pmod{n}$ *imply* $a + c \equiv b + d \pmod{n}$
7. $a \equiv b \pmod{n}$ *and* $c \equiv d \pmod{n}$ *imply* $ac \equiv bd \pmod{n}$



4.5.1 Turing's Code (Version 2.0)

we had the basic idea right (v1 : multiply the message by the key), but erred in using conventional arithmetic instead of modular arithmetic.

Beforehand The sender and receiver agree on a large prime p, which may be made public. (This will be the modulus for all our arithmetic.) They also agree on a secret key $k \in \{1, 2, \ldots, p-1\}$.

Encryption The message m can be any integer in the set $\{0, 1, 2, \ldots, p-1\}$; in particular, the message is no longer required to be a prime. The sender encrypts the message m to produce m^* by computing:

$$m^* = \text{rem}(mk, p) \tag{4.7}$$

4.6 Arithmetic with a Prime Modulus

4.6.1 Multiplicative Inverses

The *multiplicative inverse* of a number x is another number x^{-1} such that:

$$x \cdot x^{-1} = 1$$

Generally, multiplicative inverses exist over the real numbers. For example, the multiplicative inverse of 3 is $1/3$ since:

$$3 \cdot \frac{1}{3} = 1$$

The sole exception is that 0 does not have an inverse.

On the other hand, inverses generally do not exist only over the integers. For example, 7 can not be multiplied by another integer to give 1.

multiplicative inverse modulo a prime number

Surprisingly, multiplicative inverses do exist when we're working *modulo a prime number*. For example, if we're working modulo 5, then 3 is a multiplicative inverse of 7, since:

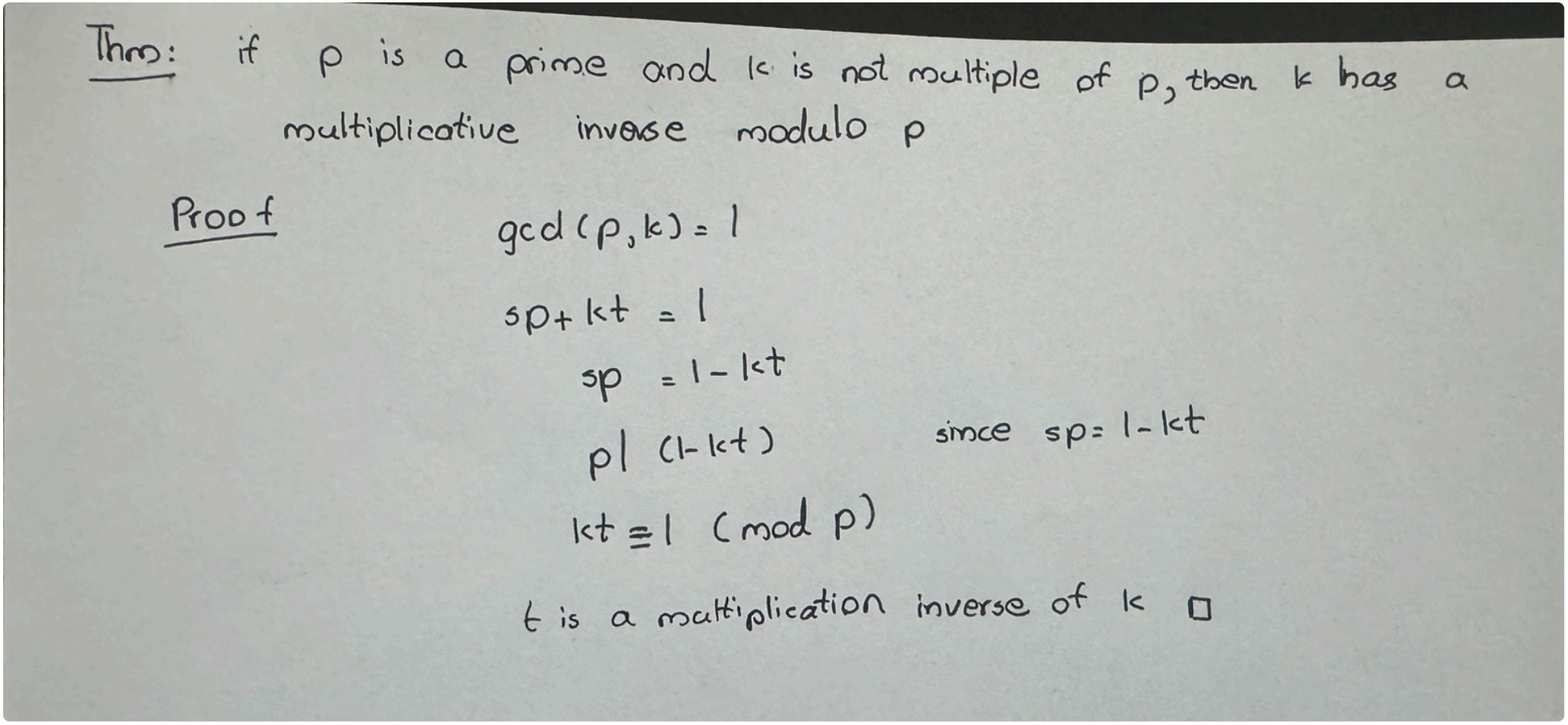
$$7 \cdot 3 \equiv 1 \pmod 5$$

(All numbers congruent to 3 modulo 5 are also multiplicative inverses of 7; for example, $7 \cdot 8 \equiv 1 \pmod 5$ as well.) The only exception is that numbers congruent to 0 modulo 5 (that is, the multiples of 5) do not have inverses, much as 0 does not have an inverse over the real numbers. Let's prove this.

> so zero and multiple of p never have multiplicative inverse modulo p when p ::= prime

Lemma 4.6.1 (prime & multiplicative inverse)

Lemma 4.6.1. If p is prime and k is not a multiple of p, then k has a multiplicative inverse modulo p.



Decryption in Turing's code (Version 2.0)

Multiplicative inverses are the key to decryption in Turing's code. Specifically, we can recover the original message by multiplying the encoded message by the *inverse* of the key:

$$\begin{aligned} m^* \cdot k^{-1} &= \text{rem}(mk, p) \cdot k^{-1} && \text{(the def. (4.7) of } m^*) \\ &\equiv (mk)k^{-1} \pmod{p} && \text{(by Cor. 4.5.2)} \\ &\equiv m \pmod{p}. \end{aligned}$$

This shows that m^*k^{-1} is congruent to the original message m. Since m was in the range $0, 1, \ldots, p-1$, we can recover it exactly by taking a remainder:

$$m = \text{rem}(m^*k^{-1}, p).$$

So all we need to decrypt the message is to find a value of k^{-1}. From the proof of Lemma 4.6.1, we know that t is such a value, where $sp + tk = 1$. Finding t is easy using the Pulverizer.

4.6.2 Cancellation

multiplicative terms can not be canceled is in congruences (modular arithmetic) differ from ordinary equations.

$$2 \times 3 \equiv 4 \times 3 \pmod{6} \neq 2 \equiv 4 \pmod{6}$$

However, this difference goes away if we're working modulo a prime; then cancellation is valid.

Lemma 4.6.2

Lemma 4.6.2. Suppose p is a prime and k is not a multiple of p. Then

$$ak \equiv bk \pmod{p} \qquad \text{implies} \qquad a \equiv b \pmod{p}$$

Proof. Multiply both sides of the congruence by k

Corollary 4.6.3

Corollary 4.6.3. *Suppose p is a prime and k is not a multiple of p. Then the sequence:*

$$\text{rem}((1 \cdot k), p), \quad \text{rem}((2 \cdot k), p), \quad \ldots, \quad \text{rem}(((p-1) \cdot k), p)$$

is a permutation[4] *of the sequence:*

$$1, \quad 2, \quad \ldots, \quad (p-1).$$

Proof. The sequence of remainders contains $p-1$ numbers. Since $i \cdot k$ is not divisible by p for $i = 1, \ldots p-1$, all these remainders are in the range 1 to $p-1$ by the definition of remainder. Furthermore, the remainders are all different: no two numbers in the range 1 to $p-1$ are congruent modulo p, and by Lemma 4.6.2, $i \cdot k \equiv j \cdot k \pmod{p}$ if and only if $i \equiv j \pmod{p}$. Thus, the sequence of remainders must contain *all* of the numbers from 1 to $p-1$ in some order. ■

4.6.3 Fermat's Little Theorem

Theorem 4.6.4 (Fermat's Little Theorem). *Suppose p is a prime and k is not a multiple of p. Then:*

$$k^{p-1} \equiv 1 \pmod{p}$$

Proof. We reason as follows:

$$\begin{aligned}
(p-1)! &::= 1 \cdot 2 \cdots (p-1) \\
&= \operatorname{rem}(k, p) \cdot \operatorname{rem}(2k, p) \cdots \operatorname{rem}((p-1)k, p) && \text{(by Cor 4.6.3)} \\
&\equiv k \cdot 2k \cdots (p-1)k \pmod{p} && \text{(by Cor 4.5.2)} \\
&\equiv (p-1)! \cdot k^{p-1} \pmod{p} && \text{(rearranging terms)}
\end{aligned}$$

Now $(p-1)!$ is not a multiple of p because the prime factorizations of $1, 2, \ldots, (p-1)$ contain only primes smaller than p. So by Lemma 4.6.2, we can cancel $(p-1)!$ from the first and last expressions, which proves the claim. ■

find inverses using Fermat's Theorem

Suppose p is a prime and k is not a multiple of p. Then, by Fermat's Theorem, we know that:

$$k^{p-1} \equiv 1 \pmod{p}$$

$$k^{p-2} \times k \equiv 1 \pmod{p}$$

Therefore, k^{p-2} must be a multiplicative inverse of k.

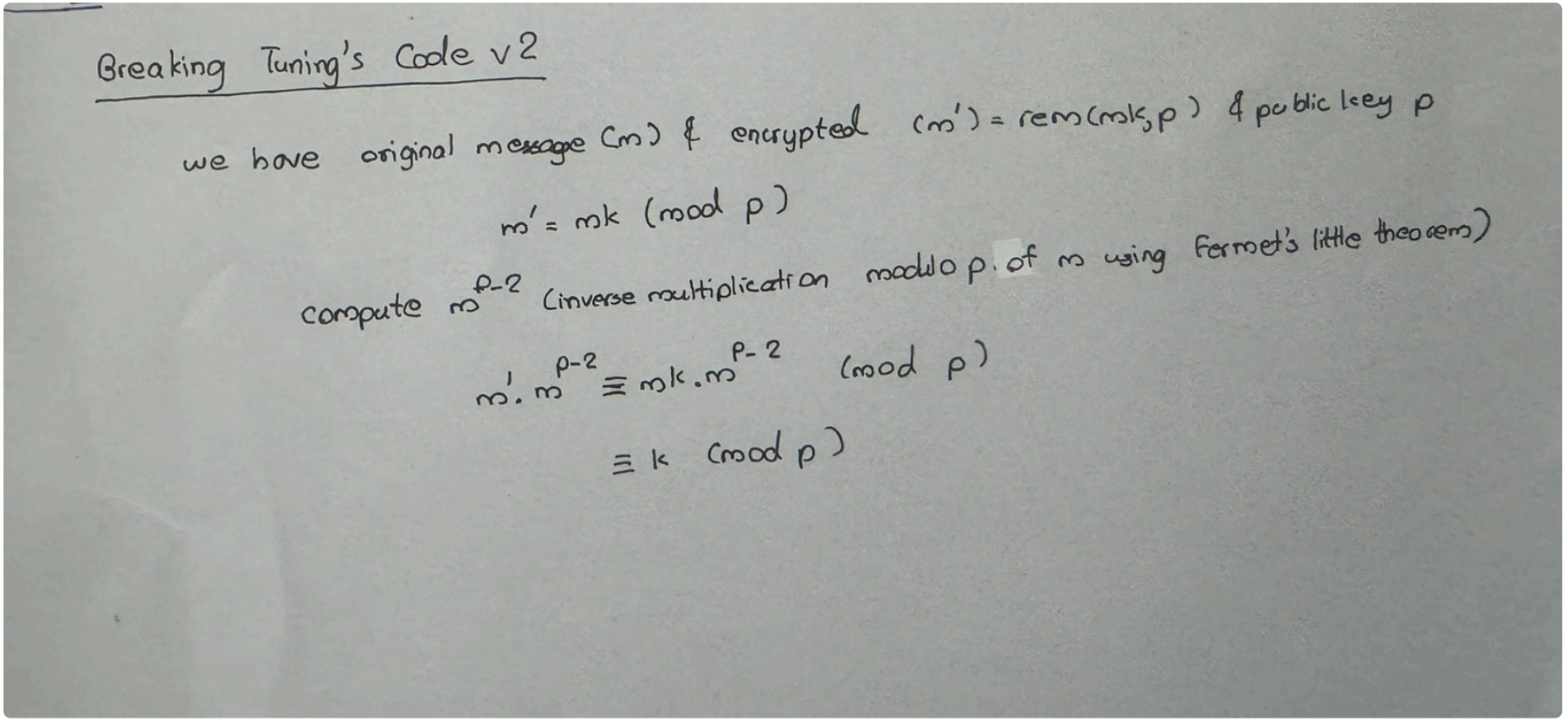
example : the multiplicative inverse of 6 modulo 17

we need to compute rem$(6^{15}, 17)$

$$
\begin{aligned}
6^2 &\equiv 36 \equiv 2 \\
6^4 &\equiv (6^2)^2 \equiv 2^2 \equiv 4 \\
6^8 &\equiv (6^4)^2 \equiv 4^2 \equiv 16 \\
6^{15} &\equiv 6^8 \cdot 6^4 \cdot 6^2 \cdot 6 \equiv 16 \cdot 4 \cdot 2 \cdot 6 \equiv 3
\end{aligned}
$$

4.6.4 Breaking Turing's Code—Again

with known-plaintext



4.7 Arithmetic with an Arbitrary Modulus

In 1977, Ronald Rivest, Adi Shamir, and Leonard Adleman at MIT proposed a highly secure cryptosystem (called RSA) based on number theory. Despite decades of attack, no significant weakness has been found.

unlike previous one, the receiver has both a secret key, which she guards closely, and a public key, which she distributes as widely as possible

The sender then encrypts his message using her widelydistributed public key.
Then she decrypts the received message using her closelyheld private key.

4.7.1 Relative Primality

this is transferring from mod prime to mod arbitrary number which still gcd is 1

- **On a Prime Modulus (p):** Because p is a prime number, **every single number** smaller than p (except 0 and multiples of p) automatically shares a greatest common divisor of 1 with it. That's why inverses are so easy to find on a prime clock.
- **On an Arbitrary Modulus (n):** Because n is composite (like 12 or 15), it has a lot of factors hiding inside it. If your number k shares *any* of those factors, $\gcd(k, n)$ becomes greater than 1, meaning an inverse is impossible. But the moment you pick a number k that shares **no factors** with n (meaning $\gcd(k, n) = 1$, like 8 and 15), it becomes just as well-behaved as a number on a prime clock!

Lemma 4.7.1 (inverse under mod a postive integer)

Lemma 4.7.1. *Let n be a positive integer. If k is* relatively prime *to n, then there exists an integer k^{-1} such that:*

$$k \cdot k^{-1} \equiv 1 \pmod{n}$$

As a consequence of this lemma, we can cancel a multiplicative term from both sides of a congruence if that term is relatively prime to the modulus:

Corollary 4.7.2. *Suppose n is a positive integer and k is relatively prime to n. If*

$$ak \equiv bk \pmod{n}$$

then

$$a \equiv b \pmod{n}$$

This holds because we can multiply both sides of the first congruence by k^{-1} and simplify to obtain the second.

The following lemma is the natural generalization of Corollary 4.6.3.

Lemma 4.7.3 (generalization of Corollary 4.6.3)

Lemma 4.7.3. *Suppose n is a positive integer and k is relatively prime to n. Let $k_1, \ldots, k_r$ denote all the integers relatively prime to n in the range 1 to $n-1$. Then the sequence:*

$$\text{rem}(k_1 \cdot k, n), \quad \text{rem}(k_2 \cdot k, n), \quad \text{rem}(k_3 \cdot k, n), \quad \ldots \quad , \text{rem}(k_r \cdot k, n)$$

is a permutation of the sequence:

$$k_1, \quad k_2, \quad \ldots \quad , k_r.$$

Lemma 4.7.3 if n is a positive integer and k is relatively prime to n.
Let $k_1, k_2, \ldots, k_r \in [1, 2, \ldots, n-1]$ denote the integer relatively prime to n $(r = \phi(n))$

Then $\underbrace{\{\text{rem}(k_1 \cdot k, n), \ldots, \text{rem}(k_r \cdot k, n)\}}_{① \ \# = r} = \underbrace{\{k_1, \ldots, k_r\}}_{③ \ C}$

Pf①: Assume $\text{rem}(k_a \cdot k, n) = \text{rem}(k_b \cdot k, n)$

$$k_a \cdot k \equiv k_b \cdot k \pmod{n}$$

$$k_a \equiv k_b \pmod{n} \text{ by corollary 4.2}$$

$$k_a = k_b \text{ since they both are between 1 and } n-1$$

therefore $\{\text{rem}(k_1 \cdot k, n), \ldots, \text{rem}(k_r \cdot k, n)\}$ set is all different with # of r

Pf②: $\gcd(n, \text{rem}(k_i \cdot k, n)) = \gcd(k_i \cdot k, n)$ $\gcd(a,b) = \gcd(b, \text{rem}(a,b))$ by Lemma 4.24. part (5)

$= 1$ by lema 4.24 part (3)

$1 = \gcd(k_i, n) = 1 \wedge \gcd(k, n) \Rightarrow \gcd(k_i k, n) = 1$

Since $\text{rem}(k_i \cdot k, n)$ will be in $0 \leq \text{rem}(k_i \cdot k, n) < n-1$ & it also relatively prime to n,

by Pf① & pf② they must be same

4.7.2 Euler's Theorem

RSA relies heavily on a generalization of Fermat's Theorem known as Euler's Theorem.

Euler's totient function

Def : $\phi(n)$ denotes the number of integers in {1,..,n} that are relatively prime to n.

If n is prime: $\phi(n) = n - 1$ (because every number less than a prime is relatively prime to it).

If n is composite: we will use following theorem

Theorem 4.7.4 (Euler's totient function for non-prime)

Theorem 4.7.4. *For any number n, if $p_1, p_2, \ldots, p_j$ are the (distinct) prime factors of n, then*

$$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\ldots\left(1 - \frac{1}{p_j}\right).$$

For example,

$$\begin{aligned}\phi(300) &= \phi(2^2 \cdot 3 \cdot 5^2) \\ &= 300\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) \\ &= 300\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{4}{5}\right) \\ &= 80.\end{aligned}$$

Corollary 4.7.5 (Euler's totient function for multiple of two prime)

Let $n = pq$ where p and q are different primes. Then $\phi(n) = (p-1)(q-1)$

Theorem 4.7.6 (Euler’s Theorem)

Theorem 4.7.6 (Euler's Theorem)**.** *Suppose n is a positive integer and k is relatively prime to n. Then*

$$k^{\phi(n)} \equiv 1 \pmod{n}$$

Proof. Let $k_1, \ldots, k_r$ denote all integers relatively prime to n such that $0 \leq k_i < n$. Then $r = \phi(n)$, by the definition of the function ϕ. The remainder of the proof mirrors the proof of Fermat's Theorem. In particular,

$$\begin{aligned}
&k_1 \cdot k_2 \cdots k_r \\
&\quad = \operatorname{rem}(k_1 \cdot k, n) \cdot \operatorname{rem}(k_2 \cdot k, n) \cdots \operatorname{rem}(k_r \cdot k, n) && \text{(by Lemma 4.7.3)} \\
&\quad \equiv (k_1 \cdot k) \cdot (k_2 \cdot k) \cdot \cdots \cdot (k_r \cdot k) \pmod{n} && \text{(by Cor 4.5.2)} \\
&\quad \equiv (k_1 \cdot k_2 \cdots k_r) \cdot k^r \pmod{n} && \text{(rearranging terms)}
\end{aligned}$$

Part (3) of Lemma 4.2.4. implies that $k_1 \cdot k_2 \cdots k_r$ is relatively prime to n. So by Corollary 4.7.2, we can cancel this product from the first and last expressions. This proves the claim. ■

We can find multiplicative inverses using Euler's theorem as we did with Fermat's theorem: if k is relatively prime to n, then $k^{\phi(n)-1}$ is a multiplicative inverse of k modulo n. However, this approach requires computing $\phi(n)$. Computing $\phi(n)$ is easy (using Theorem 4.7.4) if we know the prime factorization of n. Unfortunately, finding the factors of n can be hard to do when n is large and so the Pulverizer is often the best approach to computing inverses modulo n.

4.8 The RSA Algorithm

The RSA Cryptosystem

Beforehand The receiver creates a public key and a secret key as follows.

1. Generate two distinct primes, p and q. Since they can be used to generate the secret key, they must be kept hidden.
2. Let $n = pq$.
3. Select an integer e such that $\gcd(e, (p-1)(q-1)) = 1$.
   The *public key* is the pair (e, n). This should be distributed widely.
4. Compute d such that $de \equiv 1 \pmod{(p-1)(q-1)}$. This can be done using the Pulverizer.
   The *secret key* is the pair (d, n). This should be kept hidden!

Encoding Given a message m, the sender first checks that $\gcd(m, n) = 1$.[a] The sender then encrypts message m to produce m' using the public key:

$$m' = \text{rem}(m^e, n).$$

Decoding The receiver decrypts message m' back to message m using the secret

key:

$$m = \text{rem}((m')^d, n).$$

[a]It would be very bad if $\gcd(m, n)$ equals p or q since then it would be easy for someone to use the encoded message to compute the secret key If $\gcd(m, n) = n$, then the encoded message would be 0, which is fairly useless. For very large values of n, it is extremely unlikely that $\gcd(m, n) \neq 1$. If this does happen, you should get a new set of keys or, at the very least, add some bits to m so that the resulting message is relatively prime to n.

Why is RSA Secure?

The entire security of RSA relies on one hard mathematical truth mentioned at the very beginning of the lecture: It is brutally hard to factor a massive number back into its original primes.

- If an attacker intercepts your public key (e, n), they know n, but they *do not* know p and q.
- To find your secret key d, they would have to factor n back into p and q. If n is a number with hundreds of digits, even the world's most powerful supercomputers would take thousands of years to factor it.
- it is safe unless they don't get your d

proving decryption bring encryption

Encryption: $m' = \text{rem}(m^e, n)$

Decryption: $m = \text{rem}((m')^d, n)$

$m' = \text{rem}(m^e, n) \rightarrow \text{rem}(m', n) \equiv m^e \pmod{n}$

$m' \equiv m^e \pmod{n}$ ②

$(m')^d \equiv m^{ed} \pmod{n}$

$\exists r \; ed = 1 + r(p-1)(q-1)$ from $de \equiv 1 \pmod{(p-1)(q-1)}$

$(m')^d = m \cdot m^{r(p-1)(q-1)} \pmod{n}$

$\gcd(m, n) = 1$ & by Euler's theorem

$m^{\phi(n)} \equiv 1 \pmod{n}$

$n = p \cdot q$ (mutiple of two distintct prime) by Corollary 9.7.5

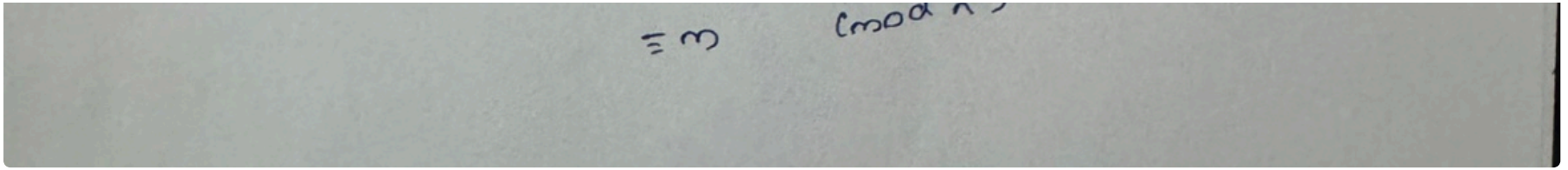
$\phi(n) = (p-1)(q-1)$

$(m')^d = m \cdot m^{r(p-1)(q-1)} \pmod{n}$

$= m \cdot (m^{(p-1)(q-1)})^r \pmod{n}$

$\equiv m \cdot 1^r \pmod{n}$

□



Recitation : Exponentiation and Modular Arithmetic

we will use this one

$$\begin{aligned}\operatorname{rem}(a \cdot b, c) &\equiv a \cdot b \pmod{c} \\ &\equiv \operatorname{rem}(a, c) \cdot \operatorname{rem}(b, c) \pmod{c} \\ &= \operatorname{rem}((\operatorname{rem}(a, c) \cdot \operatorname{rem}(b, c)), c)\end{aligned}$$

Find rem(23 · 61 · 19, 17)

We could find the remainder of 23 · 61 · 19 = 26657 divided by 17, but that would be a lot of unnecessary work!
Instead, we notice the fact that 23 ≡ 6 (mod 17), 61 ≡ 10 (mod 17), and 19 ≡ 2 (mod 17).
Therefore, 23 · 61 · 19 ≡ 6 · 10 · 2 (mod 17).

Fast Modular Exponentiation Guide

Fast Modular Exponentiation (Repeated Squaring)

How It Works

1. **binary expansion** : the binary expansion of 149 is 10010101,
2. **Repeatedly Square**: starting from power 1 to maximum power find the mod(n) for each
3. **Combine Powers**: Multiply together only the specific powers that correspond to the 1s in your binary expansion.

Example: Encrypt a message $m = 5$ with $e = 149$ and $n = 17$.

Note that the binary expansion of 149 is 10010101, so one can compute $\text{rem}\,(5^{149}, 17)$ by computing $\text{rem}\,(5^{128+16+4+1}, 17)$.

$$\begin{aligned} 5^2 &\equiv 8 \pmod{17} \\ 5^4 &\equiv 8 \cdot 8 \equiv 13 \pmod{17} \\ 5^8 &\equiv 13 \cdot 13 \equiv 16 \pmod{17} \\ 5^{16} &\equiv 16 \cdot 16 \equiv 1 \pmod{17} \\ 5^{32} &\equiv 1 \cdot 1 \equiv 1 \pmod{17} \\ 5^{64} &\equiv 1 \cdot 1 \equiv 1 \pmod{17} \\ 5^{128} &\equiv 1 \cdot 1 \equiv 1 \pmod{17} \end{aligned}$$

$\text{rem}(5^{16\cdot 9+4+1}, 17) \equiv (5^{16})^9 \cdot 5^4 \cdot 5 \equiv 1^9 \cdot 13 \cdot 5 \equiv 14 \pmod{17}$

we get 14 easily

problem set

What is a square modulo (p)?

A number (n) is called a square modulo (p) if there exists an integer (x) such that:

$$\boxed{n \equiv x^2 \pmod{p}}$$

In simple words:

1. Choose an integer (x).
2. Square it (x^2).

3. Divide the result by (p).
4. Look at the remainder.

If n is that remainder (modulo p), then (n) is a square modulo (p).

> **Note**
>
> we only need to check 0 to $p-1$.

Theorem 1 (Euler's Criterion)

$$1.\ \textit{If } n \textit{ is a square modulo } p \textit{ then } n^{\frac{p-1}{2}} \equiv 1 \pmod{p}.$$

$$2.\ \textit{If } n \textit{ is not a square modulo } p \textit{ then } n^{\frac{p-1}{2}} \equiv -1 \pmod{p}.$$